Example 6.4 Prove that $\sec\theta - \cos\theta = \tan\theta \sin\theta$

Solution $\sec\theta - \cos\theta = \dfrac{1}{\cos\theta} - \cos\theta = \dfrac{1-\cos^2\theta}{\cos\theta}$

$= \dfrac{\sin^2\theta}{\cos\theta}$ $\left[\because 1-\cos^2\theta = \sin^2\theta\right]$

$= \dfrac{\sin\theta}{\cos\theta} \times \sin\theta = \tan\theta\sin\theta$

Example 6.5 Prove that $\sqrt{\dfrac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$

Solution $\sqrt{\dfrac{1+\cos\theta}{1-\cos\theta}} = \sqrt{\dfrac{1+\cos\theta}{1-\cos\theta} \times \dfrac{1+\cos\theta}{1+\cos\theta}}$ [multiply numerator and denominator by the conjugate of $1-\cos\theta$]

$= \sqrt{\dfrac{(1+\cos\theta)^2}{1-\cos^2\theta}} = \dfrac{1+\cos\theta}{\sqrt{\sin^2\theta}}$ $\left[\because \sin^2\theta + \cos^2\theta = 1\right]$

$= \dfrac{1+\cos\theta}{\sin\theta} = \operatorname{cosec}\theta + \cot\theta$

Example 6.6 Prove that $\dfrac{\sec\theta}{\sin\theta} - \dfrac{\sin\theta}{\cos\theta} = \cot\theta$

Solution $\dfrac{\sec\theta}{\sin\theta} - \dfrac{\sin\theta}{\cos\theta} = \dfrac{\dfrac{1}{\cos\theta}}{\sin\theta} - \dfrac{\sin\theta}{\cos\theta} = \dfrac{1}{\sin\theta\cos\theta} - \dfrac{\sin\theta}{\cos\theta}$

$= \dfrac{1-\sin^2\theta}{\sin\theta\cos\theta} = \dfrac{\cos^2\theta}{\sin\theta\cos\theta} = \cot\theta$

Example 6.7 Prove that $\sin^2 A\cos^2 B + \cos^2 A\sin^2 B + \cos^2 A\cos^2 B + \sin^2 A\sin^2 B = 1$

Solution $\sin^2 A\cos^2 B + \cos^2 A\sin^2 B + \cos^2 A\cos^2 B + \sin^2 A\sin^2 B$

$= \sin^2 A\cos^2 B + \sin^2 A\sin^2 B + \cos^2 A\sin^2 B + \cos^2 A\cos^2 B$

$= \sin^2 A(\cos^2 B + \sin^2 B) + \cos^2 A(\sin^2 B + \cos^2 B)$

$= \sin^2 A(1) + \cos^2 A(1)$ $(\because \sin^2 B + \cos^2 B = 1)$

$= \sin^2 A + \cos^2 A = 1$

Example 6.8 If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, then prove that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$

Solution Now, $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$.

Squaring both sides,

$(\cos\theta + \sin\theta)^2 = (\sqrt{2}\cos\theta)^2$

$\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta = 2\cos^2\theta$

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